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Department of Computer Engineering
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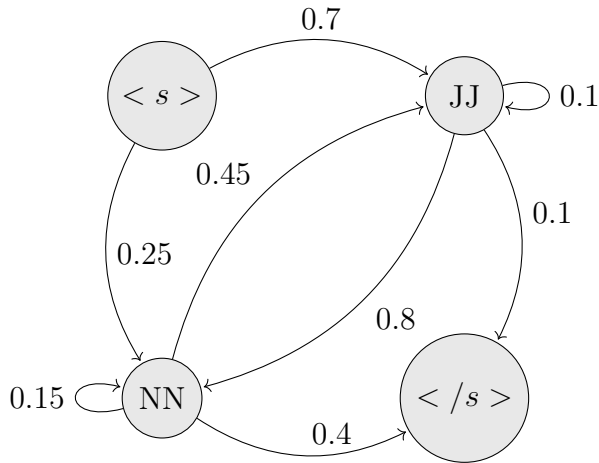
(CSDC7013) Natural Language Processing

S O L U T I O N to Internal Assessment 1

1. Attempt any **TWO** from a, b and c **AND** any **THREE** from d, e, f and g

(a) [02 Marks] With the given HMM, calculate the probability that the sentence “hot lead cold feet” will be assigned the parts of speech; JJ NN JJ NN

Values on the edges are transition probabilities and the table below gives emission probabilities:

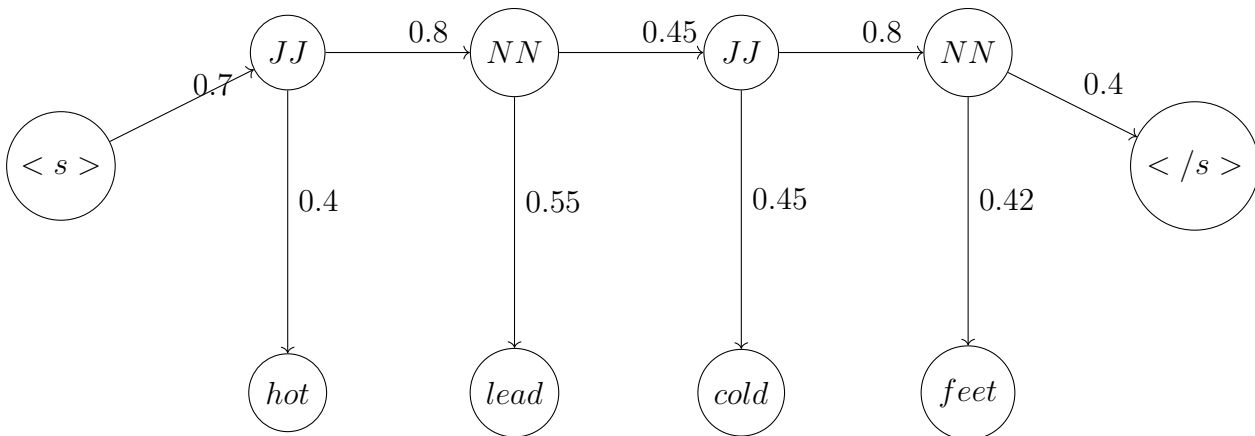


	hot	lead	cold	feet
JJ	0.4	0.1	0.45	0.05
NN	0.01	0.55	0.02	0.42

Rubrics:

- [02 Marks] Correct model and correct computation
- [01 Marks] Partially correct model and/or computation
- [00 Marks] Wrong model. Wrong computation.

Solution:



The required probability is $= 0.7 \times 0.4 \times 0.8 \times 0.45 \times 0.45 \times 0.8 \times 0.42 \times 0.4 = 0.003353$

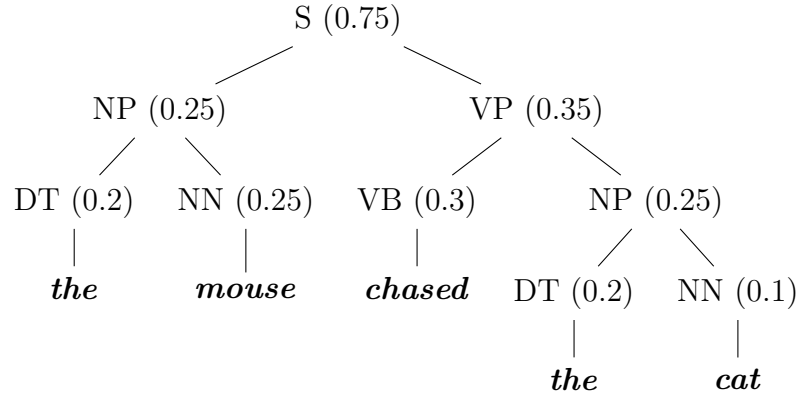
(b) [02 Marks] Compute the probability of the parse tree of the input string “The mouse chased the cat”, given the following partial PCFG

- $S \rightarrow NP VP$ 0.75
- $NP \rightarrow DT NN$ 0.25
- $VP \rightarrow VB NP$ 0.35
- $NN \rightarrow \text{mouse}$ 0.25 | **cat** 0.1 | **dog** 0.4
- $VB \rightarrow \text{chased}$ 0.3 | **dodged** 0.5
- $DT \rightarrow \text{the}$ 0.2 | **a** 0.35

Rubrics:

- [02 Marks] Correct parse tree, correct annotations and correct computation.
- [01 Marks] Partially correct Correct parse tree, and / or partially correct annotations and partially correct computation.
- [00 Marks] Completely wrong parse tree and computation

Solution:



The probability of the parse tree of the given input string “The mouse chased the cat” is $= 0.75 \times 0.25 \times 0.2 \times 0.25 \times 0.35 \times 0.3 \times 0.25 \times 0.2 \times 0.1 = 0.000004922$

(c) [02 Marks] Perform left most derivation and compute the parse tree to check the string “**The Panda eats leaves**”, belongs to the following grammar:

$S \rightarrow NP VP$

$NP \rightarrow DT NN$

$VP \rightarrow VB NN$

$DT \rightarrow a \mid the$

$NN \rightarrow \text{Man} \mid \text{Woman} \mid \text{Panda} \mid \text{Lion} \mid \text{leaves}$

$VB \rightarrow \text{eats} \mid \text{shoots} \mid \text{leaves}$

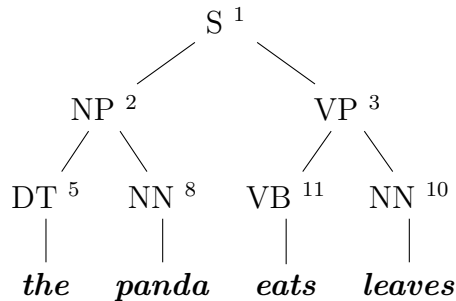
Rubrics:

- [02 Marks] Correct leftmost derivation and correct parse tree
- [01 Marks] Partially correct leftmost derivation and / or correct parse tree
- [00 Marks] Completely wrong LMD and tree

Solution:

- **Label the Grammar rules:**
 - $S \rightarrow NP VP$ ¹

- NP $\rightarrow$ DT NN ²
- VP $\rightarrow$ VB NN ³
- DT $\rightarrow$ a ⁴ | the ⁵
- NN $\rightarrow$ Man ⁶ | Woman ⁷ | Panda ⁸ | Lion ⁹ | leaves ¹⁰
- VB $\rightarrow$ eats ¹¹ | shoots ¹² | leaves ¹³
- **Perform Leftmost Derivation**
 - S $\rightarrow$ NP VP (1)
 - $\rightarrow$ DT NN VP (2)
 - $\rightarrow$ *the* NN VP (5)
 - $\rightarrow$ *the panda* VP (8)
 - $\rightarrow$ *the panda* VB NN (3)
 - $\rightarrow$ *the panda eats* NN (11)
 - $\rightarrow$ *the panda eats leaves* (10)
- **The Parse Tree**



(d) [02 Marks]

Given the following regular expression

R = "\b[A-Za-z]*[[^]aeiou]{2}[a-z]*\b"

and the following input sentence, write all words matched with regular expression *R*

Dolly on a trolley got a seat by golly. Guss on a bus got a seat without fuss.

Rubrics:

- [02 Marks] All matched words correctly identified.
- [01 Marks] Some matched words correctly identified.
- [00 Marks] No or very few matched words correctly identified.

Solution:

Words Matched: { Dolly, trolley, golly, Guss, without, fuss }

(e) [02 Marks] Write the inflectional morphemes identified in the following discourse.

Them that asked no questions, cleverer not told lies.

Rubrics:

- [02 Marks] All correct inflectional morphemes listed.
- [01 Marks] Partially correct list inflectional morphemes
- [00 Marks] Completely wrong

Solution:

Inflectional Morphemes:

- ed in "asked" (past tense)
- s in "questions" (plural)
- er in "cleverer" (comparative)
- ed in "told" (past tense)
- s in "lies" (plural)

- (f) [02 Marks] Compute the perplexity of "I like sweets", by considering the bigram model from the following text.

I like many things to eat. I like cake. I like biscuits. And, yes, sweets. I do like them. I like jam. It is like sweets.

Rubrics:

- [02 Marks] Correct computation of perplexity
- [01 Marks] Partially correct computation
- [00 Marks] Completely wrong

Solution:

- **List the bigrams in the given sentence:** I like sweets.
 - $\langle s \rangle$ I like sweets $\langle /s \rangle$
 - **Bigrams:** $\langle \langle s \rangle, I \rangle$, $\langle I, \text{like} \rangle$, $\langle \text{like}, \text{sweets} \rangle$, $\langle \text{sweets}, \langle /s \rangle \rangle$
- **Write each sentence along with start and end of sentences markers.**
 - $\langle s \rangle$ I like many things to eat. $\langle /s \rangle$
 - $\langle s \rangle$ I like cake, $\langle /s \rangle$
 - $\langle s \rangle$ I like biscuits. $\langle /s \rangle$
 - $\langle s \rangle$ And, yes, sweets. $\langle /s \rangle$
 - $\langle s \rangle$ I do like them. $\langle /s \rangle$
 - $\langle s \rangle$ I like jam. $\langle /s \rangle$
 - $\langle s \rangle$ It is like sweets. $\langle /s \rangle$
- **Compute the probabilities of the bigrams**
 - $P(I | \langle s \rangle) = \text{count}(\langle s \rangle I) / \text{count}(\langle s \rangle) = 5/7 = 0.715$
 - $P(\text{like} | I) = \text{count}(I \text{ like}) / \text{count}(I) = 4/5 = 0.8$
 - $P(\text{sweets} | \text{like}) = \text{count}(\text{like sweets}) / \text{count}(\text{like}) = 1/6 = 0.167$
 - $P(\langle /s \rangle | \text{sweets}) = \text{count}(\text{sweets} \langle /s \rangle) / \text{count}(\text{sweets}) = 2/2 = 1$
- **Compute perplexity of the given sentence:**

$$PP(W) = \sqrt[3]{\frac{1}{P(W)}}$$

Where $P(W) = 0.715 \times 0.8 \times 0.167 \times 1 = 0.095524$

$$PP(W) = \sqrt[3]{\frac{1}{0.095524}} = 2.188$$

- (g) [02 Marks] Compute the measure of the following words, with respect to Porter Stemmer:
Pyrotechnique, acronymania, gyroscope, pylorus

Rubrics:

- [02 Marks] Correct measure computed for all words.
- [01 Marks] Partially correct measure for few words.
- [00 Marks] Completely wrong

Solution:

Find the VC pairs (Sequence and Vowel and consonant pairs)

- **Pyrotechnique** : P[yr][ot][echn][iq]ue : measure = 4
- **acronymania** : [acr][on][ym][an]ia : measure = 4
- **gyroscope** : g[yr][osc][op]e : measure = 3
- **pylorus** : p[yl][or][us] : measure = 3

2. Attempt any **ONE**

- (a) [05 Marks] Consider the following corpus

- < s > the/DT students/NN pass/V the/DT test/NN < /s >
- < s > the/DT students/NN wait/V for/P the/DT result/NN < /s >
- < s > teachers/NN test/V students/NN < /s >

Compute the emission and transition probabilities for a bigram HMM

Rubrics:

- [05-04 Marks] Correct computation of emission and transition probabilities
- [03-02 Marks] Partially correct computation of emission and transition probabilities
- [01-00 Marks] Completely wrong

Solution:

Using the formula:

$$P(t_i|t_{i-1}) = \frac{\text{count}(t_{i-1} t_i)}{\text{count}(t_{i-1})} \quad P(w_i|t_i) = \frac{\text{count}(t_i w_i)}{\text{count}(t_i)}$$

In the given corpus, the tags are { DT, NN, V, P }

And the words (vocabulary/lexicon) are (in small case arranged in alphabetical order

{ for, pass, result, students, teachers, test, the, wait }

Emission Probabilities: Matrix B

$$B = \begin{bmatrix} & for & pass & result & students & teachers & test & the & wait \\ DT & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ NN & 0 & 0 & 1/6 & 1/2 & 1/6 & 1/6 & 1 & 0 \\ V & 0 & 1/3 & 0 & 0 & 0 & 1/3 & 0 & 1/3 \\ P & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Transition Probabilities: Matrix A

$$A = \begin{bmatrix} & DT & NN & V & P & </s> \\ <s> & 2/3 & 1/3 & 0 & 0 & 0 \\ DT & 0 & 1 & 0 & 0 & 0 \\ NN & 0 & 0 & 1/2 & 0 & 1/2 \\ V & 1/3 & 1/3 & 0 & 1/3 & 0 \\ P & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- (b) [05 Marks] Consider the grammar of Q. 1. (c) and compute the CKY table to check the string "The woman shoots Lion", belongs to the given grammar

Rubrics:

- [05-04 Marks] Correct computation of CKY table
- [03-02 Marks] Partially correct computation of CKY table
- [01-00 Marks] Completely wrong

Solution:

The CKY table is:

$$\begin{bmatrix} the & woman & shoots & lion \\ DT & NP & & S \\ & NN & & \\ & & VB & VP \\ & & & NN \end{bmatrix}$$

With the start state S coming in the top right cell, we conclude that the given string belongs to the grammar.

3. Attempt any ONE

- (a) [05 Marks] Exemplify the challenges in natural language processing.

Rubrics:

- [05-04 Marks] Correct challenges listed with examples
- [03-02 Marks] Partially correct list and/or irrelevant examples
- [01-00 Marks] Completely wrong

Solution:

- (b) [05 Marks] Summarize three applications of NLP

Rubrics:

- [05-04 Marks] Correct summarization of applications of NLP
- [03-02 Marks] Partially correct summarization
- [01-00 Marks] Completely wrong

Solution: